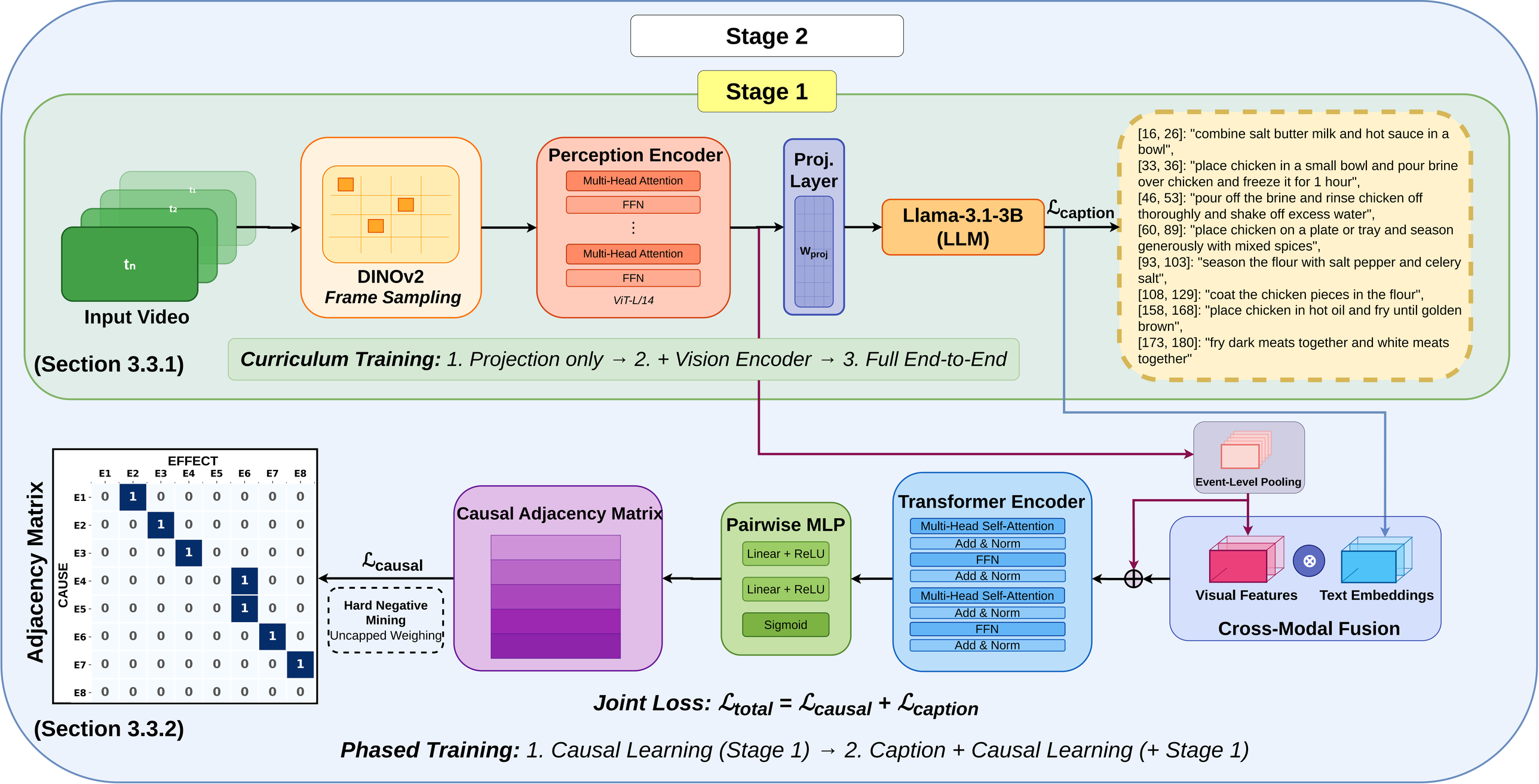

Stage 2
Stage 1
Input Video
DINOv2
Frame Sampling
Perception Encoder
Multi-Head Attention
FFN
Multi-Head Attention
FFN
ViT-L/14
Proj. Layer
Llama-3.1-3B (LLM)
$\mathcal{L}_{caption}$
[16, 26]: "combine salt butter milk and hot sauce in a bowl",
[33, 36]: "place chicken in a small bowl and pour brine over chicken and freeze it for 1 hour",
[46, 53]: "pour off the brine and rinse chicken off thoroughly and shake off excess water",
[60, 89]: "place chicken on a plate or tray and season generously with mixed spices",
[93, 103]: "season the flour with salt pepper and celery salt",
[108, 129]: "coat the chicken pieces in the flour",
[158, 168]: "place chicken in hot oil and fry until golden brown",
[173, 180]: "fry dark meats together and white meats together"
(Section 3.3.1)
Curriculum Training: 1. Projection only → 2. + Vision Encoder → 3. Full End-to-End
Event-Level Pooling
Visual Features
Text Embeddings
Cross-Modal Fusion
Transformer Encoder
Multi-Head Self-Attention
Add & Norm
FFN
Add & Norm
Multi-Head Self-Attention
Add & Norm
FFN
Add & Norm
Pairwise MLP
Linear + ReLU
Linear + ReLU
Sigmoid
Causal Adjacency Matrix
$\mathcal{L}_{causal}$
Hard Negative Mining
Uncapped Weighing
Adjacency Matrix
EFFECT
CAUSE
E1 E2 E3 E4 E5 E6 E7 E8
(Section 3.3.2)
Joint Loss: $\mathcal{L}_{total} = \mathcal{L}_{causal} + \mathcal{L}_{caption}$
Phased Training: 1. Causal Learning (Stage 1) → 2. Caption + Causal Learning (+ Stage 1)